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ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА

Достигаем ли мы лучших результатов, находясь в состоянии стресса? Эмпирическое исследование на российских данных

Бакалаврская программа

Программа Бакалавр экономики

Совместная программа по экономике НИУ ВШЭ и РЭШ

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Do we perform better when under stress? Empirical evidence from Russia

Abstract

Согласно многочисленным исследованиям, образовательная среда в России является одной из самых конкурентных в мире, вследствие чего российские студенты испытывают высокие уровни стресса и отмечают худшее психологическое состояние по сравнению с подростками из Европы и США. В своей работе я исследую влияние продолжительного стресса на академическую успеваемость школьников в условиях, когда цена ошибки очень высока. Я рассматриваю положительный и отрицательный механизмы воздействия, используя неожиданную отмену заключительного этапа Всероссийской Олимпиады Школьников по экономике в 2020 году как естественный эксперимент, позволяющий сравнить уровни стресса и академические показатели участников Олимпиады разных лет. Применяя метод "Разницы разницей"на самостоятельно собранных данных, я нахожу, что школьники, лишенные участия в Олимпиаде в 2020 году, показывают значительно более высокие результаты по сравнению с другими школьниками, участвующими повторно, что свидетельствует о преобладании позитивного эффекта над негативным. Следующим шагом я использую наличие домашнего животного в качестве инструментальной переменной для уровня испытанного стресса во время карантина 2020 года, и нахожу, что негативное влияние стресса на академическую успеваемость является незначительным и статистически неотличимым от нуля.

Abstract

Numerous studies identify the Russian educational system as one of the most rigorous and competitive worldwide. Consequently, Russian adolescents frequently experience intense stress, often resulting in poorer average psychological well-being. This study seeks to examine whether sustained stress enhances or impairs academic performance when the perceived stakes are exceptionally high. To address this question, I analyze a positive and negative channels of influence, using an unexpected cancellation of the 2020 national stage of the All-Russian Economics Olympiad as a natural experiment to compare stress and performance across year cohorts. Using the Difference-in-Differences approach on self-collected data, I find that affected participants performed significantly better than previous cohorts. This suggests that the positive effect outweighed the negative one. Subsequently, a Two-Stage Least Squares (2SLS) approach is employed, utilizing pet ownership as a instrumental variable to isolate the distinct effects of stress. The results suggest that the adverse impact of stress on academic performance is negligible and statistically insignificant.

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1 Introduction

"Out of life's school of war — what doesn't kill me, makes me stronger..." — Friedrich Nietzsche, "Twilight of the Idols" (1888).

According to numerous studies, Russian students face a much heavier academic load than their Western peers. For instance, for Economic Co-operation and Development (2016) reports that 15-year-olds in Russia spend about 9.7 hours per week doing homework, which is twice more than the OECD mean. The stakes get so high mainly because of the extreme competition for university spots. As a result, students become more susceptible to high levels of anxiety and mental disorders. Chumakov et al. (2021) interviewed St. Petersburg medical students to find out that almost 75% of them showed burnout on the Oldenburg Burnout Inventory, —far above 43% burnout rate reported by U.S. medical students.

However, it is still not clear whether such highly competitive atmosphere promote better results via raising one's motivation to study or, on the contrary, it causes exhaustion of mental resource and subsequent academic impairment. In this study, I address the ambiguity of the long-lasting influence of stress on academic achievement by collecting data on participants of intellectual competition in economics and using the Covid-19 lockdown as an external shock that caused an increase of anxiety.

Except for the lack of economic research on the topic, there are also several features that make my study valuable and unique. First, it uses a natural experiment that speaks in favor of its internal validity. Second, the reviewed literature indicates that nations with a highly competitive academic environment, such as China and Russia, might display outcomes distinct from Western countries (Ye et al. (2019)), where academic pressure tends to be less intense and stakes are not as high as in China or Russia[1]. Therefore, the results I expect to obtain will contribute to the diversity of the context, as the majority of studies in the field focus on European populations.

2 Literature Review

2.1 Why stress is related to economics?

Although the influence of stress on a human's body and behavior is widely discussed in psychological and physician literature, one may doubt why stress is also an economics-

[1] China and Russia are high-stakes country also because of high social inequality in these countries. Both in China and Russia education is one of the main social lifts that are available to low-income families.

related problem. There are two main clues for that: firstly, stress puts an enormous amount of pressure on the healthcare system of any modern country in the world. An economical burden of stress was considered in great detail by **Kessler and Greenberg (2002)** in the article "The economic burden of anxiety and stress disorders". The authors propose that, although estimated by Greenberg et al. (1999) as much as $42 billion, economic consequences of stress are indeed even higher. This is because of the great variety of ways stress can affect human health. When one is trying to estimate the economic burden of anxiety, it becomes difficult to intertwine between stress itself and other mental and physical disorders caused by stress. For instance, chronic stress affects the immune system, hence increases the risks of infections, cardiovascular disease, depression, drug addiction and other chronic diseases (Maraksina et al. 2024). This speaks in favor of a considerable underestimation of the economic burden of distress.

The most important features of stress that make it so costly are its high life-time prevalence, chronicity and early age of onset. Following the results of the Midlife Development in the US (MIDUS) survey, Kessler and Greenberg (2002) argue that distress is the fourth most prevalent disorder that interferes with one's daily functioning. In addition, it has a uniquely early age of onset, with a median age "in less than 15 years of age", which means stress disorders start to intervene in human lives from adolescence, while the rest chronic physical disorders occur much later.

In addition to all discussed above, stress is well known for its deteriorating effect on human productivity and performance. Following the National Comorbidity Survey (NCS), Kessler and Greenberg (2002) found that while stress has a significant effect on employee work cuts (defined as missing part of a day or working less efficiently than usual), no impact was established in relation to work loss (defined as missing a full day of work), which implies that stress indirectly decreases the product of the label, not through fewer working hours but through decreased efficiency. It makes this effect even harder to compute, as it is revealed in the form of opportunity costs or neglected profits, but its effects are, indeed, substantial. Taking $42 billion of the initial estimations of the annual cost of anxiety disorders (Greenberg et al., 1999) and adding up $100 billion of annual excluded costs of long-term opportunity costs, Kessler and Greenberg (2002) end up with estimating an economical burden of anxiety as high as $142 billion per year for the US, 2002. Certainly, reducing these costs is what economists should do.

2.2 Stress as a performance impairer

The impact of stress on performance (both at work and at school) is the topic that draws the attention of researchers from different fields. Articles on this theme are published in psychological, medical journals, journals of human resources and social sciences, journals of management, and business. Economists may be interested in the consequences for educational achievements, since it is strictly related to subsequent school dropout, university acceptance rates, career proficiency, and income level, which are all economic indicators. I will focus next on the noneconomic articles that use valid research design due to the scarcity of econometrics analysis in this topic: although surveys in this topic are indeed abundant, the majority of them use correlation or regression analysis that does not actually lead to causal link between two variables. Furthermore, since the link between stress and performance (either in the workplace or academic settings) might not be straightforward, as we will explore below, the common approaches employed in most of the examined studies are not entirely dependable. All of the discussed above shed light on why economical research with convincing research design is needed in this topic.

In their enlightening paper titled "Mental Health and Academic Performance: A Study on Selection and Causation Effects from Childhood to Early Adulthood", Agnafors et al. (2020) illustrate the process by which children exhibiting behavioral and emotional issues at a young age are socially selected into groups with higher risk of academic impairments, such as achieving lower grades and lacking eligibility for higher education. By so, they challenge results of numerous physiological surveys which claimed a negative relationship between emotional distress and subsequent academic failures to be causal. They also confirm results by Halpern-Manners et al. (2016) that show that education achievements do not influence mental health.

Halpern-Manners et al. (2016) in their work "The Relationship between Education and Mental Health: New Evidence from a Discordant Twin Study" provide a convincing evidence that the causal relationship between education level and distress, if presence, is one-sided. Specifically, they refuse any causal effect of educational achievements on mental health (anxiety in particular). This contradicts previously obtained results that told us the discussed causal relationship is positive as "more education generates additional social and economic resources, leading to fewer stressors, better coping strategies, and more autonomous lifestyles". Indeed, Halpern-Manners et al. 2016 proved that this link 'derives from "preexisting" differences between individuals (the selection hypothesis)', thus no causality occurs. The authors' results are highly robust and trustworthy due to their research design: they use a large

dataset of identical adult twins from the US, and get rid of genetic traits and family characteristics bias by regressing within-twin-pair differences in mental health on within-twin-pair differences in education after controlling for covariates. By so, Halpern-Manners et al. (2016) partial out any omitted variables (in fact, it is a fixed-effect regression model) to make the residual orthogonal to schooling, and thus provide an unbiased estimation of the true educational effect on mental health. They show that the true causal effect of education on mental health is zero, conditional on environmental and genetic features of respondents, which means that all previous results that stood for the presence of such a link were attributed to the preexisting respondents' characteristics, or in other words were biased. This important implementation is specifically important for my further research as it enables to deny potential reverse-causality problems.

2.3 Types of stressors that can affect academic performance

Stress is a concept that implies several types of stressors at the same time, but it is necessary to differentiate them on closer inspection. This is why I will describe the main types of stressors in relation to educational success next. Afterwards, the isolated effect of academic stress will be reviewed, as it is the type of stress that causes the most divergent results.
Chinese meta-analysis article named "A Review on the Relationship Between Chinese Adolescents' Stress and Academic Achievement" by Ye et al. (2019) proposes several main stressors that disturb highly motivated school students from China. The authors state that students face tough academic competition since early childhood as they strive to enroll at a prestigious university, although only a half of them eventually succeed. This context is especially close to my research field, as I study the effect of stress on a sample of motivated school students from Russia in a highly competitive atmosphere, therefore it may be reasonable to assume that a list of the most severe stressors is close to the one Ye et al. (2019) propose.

To be more precise, Ye et al. (2019) name the three most serious stressors for Chinese adolescents, they are academic stress, financial stress, and interpersonal stress. The last one was not shown to have any effect on academic achievements (except for teacher-student relationship issues), so financial and academic stress is what interests me more. A significant negative relationship was established between family financial stress and academic achievements of Chinese school students (Zhang (2010)). However, the link between academic stress and achievement is not so obvious. While some research stands for negative linear relationship between academic stress and academic achievement (Kaplan et al. (2005), Ye

et al. (2014)), Ye et al. (2019) argue that the relationship may be of an inverted U-shape (Middlebrooks and Audage (2008)).

2.4 Research based on Russian data

Finally, Fomina et al. (2024) have published a study that highlights Russia-specific aspects regarding the impact of academic stress on performance. Their findings corroborate earlier results by Matyushkina (2016), and (2023), who suggested that slight doses of academic stress have a beneficial impact. Essentially, prior studies depicted an inverted U-shaped correlation concerning academic performance, and this was reaffirmed by Fomina et al. (2024).

Fomina et al. (2024) define academic stress as student's psychological reaction to various learning difficulties, which they perceive as not corresponding to the individual resources (cognitive, personal) to cope with a difficult situation. They claim that exams are the most severe stressors in this context and they affect one's performance long before the exam itself via academical anxiety, especially it is true in the period of late adolescence when a teenager faces the highest uncertainty about the future (16-23 years old). Fomina et al. (2024) draw special attention to the higher prevalence of uncertainty stress in the context of influence on academic performance, especially examining post-Covid data. Despite many foreign studies, the authors also report that Russian adolescents show a significantly different relationship between academic stress and achievements: students with higher academic stress show higher academic performance, but often have a higher probability of suicide risk (Symaniuk et al. (2019)). The uniqueness of the Russian results is one more reason to conduct such a survey.

In their study Symaniuk et al. (2019), the researchers examine high school students from a prestigious Russian lyceum. They assert that students attending elite schools are subjected to increased stress due to heightened academic demands. These ambitious students often engage in numerous intellectual competitions and Olympiads, which significantly augment their distress. Interestingly, they found a positive correlation between stress levels and academic success of Russian students. Such a result sharply contrasts with international findings that typically indicate a negative relationship between these factors. It is not clear whether this positive correlation is attributed to the sampling features, selection bias or primitive instrumental tools: Spearman's correlation was chosen for the analysis, though it was already discussed above that the studied relationship may not be linear. Moreover, the majority of the respondents (76,1%) reported low or extremely low stress levels that does not coincide with the authors' assumption about increased stress levels in elite Russian schools. In addition, the percent of respondents with high or extremely high stress levels is too small

(6,6% or 13 respondents) to draw any unbiased conclusions, thus further research is needed to examine this topic once again.

Summing-up, there is a whole in the research literature examining causal effect of anxiety disorders on educational achievements, albeit it is a significant economic issue that determines one's productivity, performance and increases healthcare costs dramatically (Kessler, and Greenberg 2002). Inverse association between stress and academic success was shown by Agnafors et al. (2020), though it is not causal. The absence of reverse causal influence of academic achievements on distress has been established by Halpern-Manners et al. (2016), which eliminates the risk of reverse causality problem in the further economics analysis. Ye et al. (2019) describe differences between 3 types of stress in the academic context, and name list of obstacles that mitigate negative effect of stress. Fomina et al. (2024) name key academic stressors for Russian students, give evidence of an inverted U-shape relationship between stress and academic achievements, and poses increased importance of uncertainty stress. Symanyuk et al. (2019) finds unexpected positive correlation between the two studied variables, even though it contradicts foreign studies. However, their results cause doubts due to small sample size and primitive research analysis tools.

3 Research question

In the current situation, Russian students find themselves living and studying in the midst of significant uncertainties: the Covid-19 pandemic, the abrupt outbreak of the Russia-Ukraine war, and the resulting educational restrictions have significantly impacted their access to resources and opportunities. These restrictions include limited access to major educational platforms such as Coursera, LinkedIn, and YouTube, along with diminished chances to study abroad or win international scholarships.[2]

Such limitations heighten anxiety about the future and simultaneously raise the stakes of intellectual competitions. With fewer opportunities available, the path to academic success becomes more competitive and uncertain. Yet, the effect on performance remains ambiguous: while long-lasting stress can hinder academic preparation, the higher perceived importance of each opportunity may also enhance motivation to prepare better. In this study I strive to answer the question: which effect prevails for highly motivated Russian adolescents when stakes are high? It is important to note that I concentrate on the continuous consequences

[2] Among the most severe limitations are the exclusion of Russian applicants from the Chevening Scholarship Program in 2022 and ETH Zurich's restrictions on its Master's programs.

of stress, so that I do not measure anxiety levels during the competition itself. Instead, I focus on the permanent raise of stress preceding it beforehand.

While a detailed mathematical model elaborating on my research question is provided in the Appendix, I will now briefly outline the basic ideas. The mechanism that I am about to incorporate is presented graphically in Figure 1 and it is as follows: first, an unfavorable shock, or a stressor, occurs[3]. Stress is a body's reaction to the stressor, and this reaction process uses our body's resources, which may lead to their exhaustion. As a result, by experiencing stress one has less energy to concentrate on some other tasks such as preparation to the intellectual competition or solving a non-trivial issue. I name this effect as an effect of stress. At the same time, unfavorable events are usually unfavorable because they limit one's opportunities or make stakes higher. Therefore, higher stakes may motivate one to enhance the effort and to do better. I name this effect as a motivation effect.

In the frames of this study I also examine how individual differences in stress-resistant traits of character influence the outcomes. By investigating varying responses to academic stress among students with different level of stress resilience, this paper seeks to contribute to a more comprehensive understanding of the relationship between emotional factors and academic performance. By so, I formulate two main hypothesis:

1. H_0 : A deteriorating effect of stress prevails on a motivational effect of stress on Russian data.

2. H_0 : Magnitudes of motivation and stress effects vary across the respondents' stress-resistance traits, where a motivation effect prevails for more stress-resistant adolescents, and is obeyed by the stress effect for the less resistant students.

[3]See Section 9.2 in the Appendix for more detailed definitions.

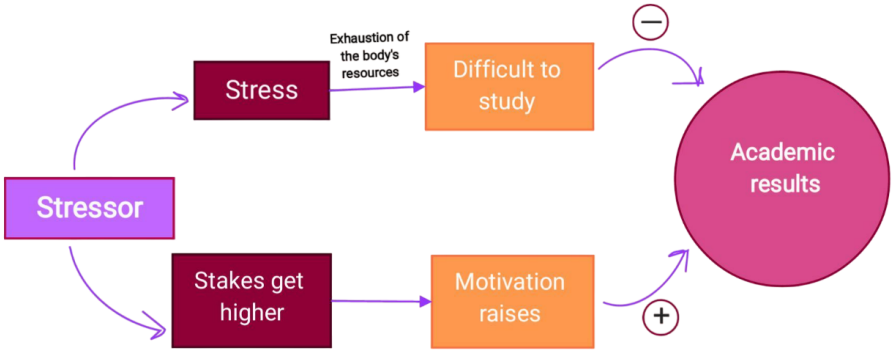


Figure 1: Mechanism of how stress influences on academic results

4 Data

4.1 Background

In this section I describe the context of data that I am using, and also comment on the necessary background. The overall idea is to use a cancellation of the final stage of the All-Russian Olympiad in economics in 2020 as an exogenous shock that triggered a lasting increase in both distress and motivation among adolescents subjected to it. I compare their results with the results of their peers not exposed to the shock and claim that the difference is attributed to the sum of stress and motivation effects. Later I try to distinguish between them using 2SLS approach.

To start with, the All-Russian Olympiad in Economics is a prestigious academic competition aimed at high school students across Russia. Participants are engaged in various rounds, starting from school-level competitions, and culminating in national finals. The complexity of the tasks increases accordingly.

The Olympiad covers a wide range of topics in economics, such as microeconomics, macroeconomics, game theory, and applied economics. It serves not only as a platform for students to showcase their knowledge and skills but it also provides opportunities to obtain scholarships for further education in economics. For instance, winners of the national final of the Olympiad receive an opportunity to enroll in any undergraduate program in economics

in Russia absolutely for free.[4] It is especially true of talented provincial school students who are willing to gain a top-ranked bachelor degree from prestigious Russian universities but cannot afford it.

Like many other academic and competitive events, the final round of All-Russian Olympiad in Economics was canceled in 2020 due to the Covid-19 pandemic restrictions. Moreover, dates of the final stage of the Olympiad were repeatedly rescheduled before it was eventually pronounced that the Olympiad would not be held that year at all. To be more precise, intra-school, municipal and regional rounds that precede the most serious national stage were successfully held before the Covid-19 pandemic in Russia, so it was only all-Russian round that was not held.

Several features of the Olympiad should be taken into account to acknowledge why cancellation of the final tours was an issue. As mentioned above, victory in the last national round provides all winners with a unique opportunity to enroll in any university for free, yet winning any previous round of the Olympiad gives nothing but the ability to compete in the subsequent tour. It is also worth noting that the privilege of enrolling in any economics university remains for three years, which means that if one wins the Olympiad[5] in the 9_{th} grade of school, they are still able to use their privilege at the end of the 11_{th} grade when enrolling at the University[6]. Though passing Unified State Exam is a requirement to complete school education, it does not play any significant role if a child has won the All-Russian Olympiad. That is, a victory in the Olympiad provides unique benefits that can bypass the usual Unified State Exam requirements.

Considering all of the above, it is clear why the national stage of the Olympiad could not be quietly canceled: the stakes for the participants were simply too high, therefore all the 11_{th}-graders who were supposed to participate in the final stage[7] received a privilege even without actually taking part in the Olympiad. It was an unprecedented case that made

[4]Higher education in Russia is paid, though there is a chance to gain a merit-based scholarship to cover the entire tuition fee. An alternative method to enroll in a prestigious Russian university at no charge is to pass Unified State Exams with flying colors. However, each Russian university sets his own requirements toward minimum scores for admission, and these requirements vary from year to year, so it is a riskier option that does not guarantee a scholarship to the most desired university.

[5]Here and further "winning the Olympiad" implies winning all four stages of the Olympiad, including the final one.

[6]School education in Russia takes 11 years to complete. Competing in All-Russian Olympiads at a national level is allowed for the 9_{th} grade school students or older. In total, there are 3 age groups that compete separately: the 9_{th}, 10_{th} and 11_{th} school grades that correspond to 15, 16 and 17 year-old cohorts. The Olympiad tasks are different for each age group to adjust to experience and technical skills, since the Olympiad is expected to reveal highly motivated and talented kids of any age.

[7]That is, those who scored high enough on the regional stage to participate in the final stage.

angry both those who planned to win a scholarship via the Unified State Exam and younger Olympiad participants[8] from the 9_{th} and 10_{th} grades. The latest ones only received the right to participate in the national stage of the Olympiad the year after it (in 2021). This was especially problematic for the 10_{th}-graders: they had to choose whether to continue preparing for the Olympiad or to give up on it and switch to preparation for the Unified State Exam, or to attempt to withstand the double load and prepare for both. Moreover, stakes became even higher as kids became elder, so that less time was left to win a scholarship.

It all converges to that such adolescents experienced a continuous raise in distress and motivation a year before they actually took part in the Olympiad in 2021. I propose to use such natural experiment to measure, which effect prevailed when stakes had gotten higher and stress increased simultaneously. I want to find out, whether deteriorating effect of stress described in Section 1 was strong enough to affect academic results of highly motivated school students.

4.2 Data Collection

My empirical research is mostly based on the self-collected data as beyond the individual Olympiad results I also needed personal characteristics and an individual assessment of stress for each respondent to use them as controls in the subsequent regression analysis. Although data containing academic results and perceived stress estimations might potentially exist elsewhere, I chose to gather data myself to incorporate an exogenous cancellation of the All-Russian Olympiad to increase internal validity of the model.

I have spread my questionnaire via three main channels: first, I published it in Telegram groups for the Olympiad participants of 2019 and 2021 years respectively. These chats unite the majority of participants and do not contain any non-participants. Second, I distributed the link with my Google survey form among NES students via our mailing system. I am very thankful to Anton Suvorov for his invaluable assistance in spreading my Google form among BAE NES 2022 - 2027 to cover as many years as possible to track a historical trend. Olympiad participants of 2019 and 2021 are BAE 2023 and BAE 2025, respectively. Finally, I used my personal set of network to spread the survey among students from other faculties: I asked my friends from the HSE Faculty of Economic Sciences to spread my questionnaire among their surrounding in order to obtain more respondents who have not won the Olympiad.

A list of questions used in the survey can be found in the Appendix. It combines personal questions about social background and living conditions during the Covid-19 lockdown with

[8] I imply those kids who were supposed to participate in the national stage, but were not allowed to.

self-reported results of the national stage of the Olympiad and a set of PSS questions to measure stress and coping abilities. PSS is a Perceived Stress Scale, a well-known self-diagnostic tool aimed to "measure feelings of lack of control and predictability as well as being overburdened..."[9]. Initially proposed by Cohen et al. (1983), The Perceived Stress Scale was invalidated in many countries and proved its efficiency. Afterwards, it was adapted for adolescents to use it at schools. I will use a short version of the PSS with 10 questions only. The version I use was proposed by Cohen (1988) and validated by Ababkov et al. (2016) for a Russian sample. Marakshina et al. (2024) and Zolotareva (2023) proved this survey set to be efficient for adolescents as well. Implying that one should not recall their psychological well-being dated 5 years back, I firstly ask a PSS-set of questions about this February[10] and next, I ask the same set of questions about Covid-19 lockdown. By doing so, I propose to use last-month stress as a reference point to make it easier estimate stress that occurred 5 years ago.

Following Marakshina et al. (2024) who proved a Russian version of PSS to have a two-factor structure, I set two sub-scales inside a PSS-questionnaire: a sub-scale of perceived self-efficacy (questions 4, 5, 7, 8, each question is worth between 0 and 4 points) and a sub-scale of perceived helplessness (questions 1, 2, 3, 6, 9, 10, each question is worth between 0 and 4 points). For each sub-scale a higher score means worse psychological well-being. A sub-scale of self-efficacy represents how successfully a respondent tackles issues and varies from 0 to 16. In other words, it measures a respondent's sense of control and coping ability (e.g. "I felt confident about my ability to handle my personal problems"). A sub-scale of helplessness is somewhat different: it assesses feelings of being overwhelmed, lost or out of control, and it ranges from 0 to 24. By adding a helplessness value and a self-efficacy value altogether, we get a final value of the respondent's PSS index, therefore the total PSS score ranges from 0 to 40.

It is important to note that the collected PSS scores are self-reported, thus they may be biased. Following Kostromina and Pisarev (2017), Russian adolescents provide unreliable estimates of their distress, and the reported results dramatically converges from the true indicators of the body measured medically. That is how they explain this phenomenon: "Возможны различные варианты обобъяснения данных результатов: школьники с трудом оценивалют свое вегетативное состояние, не могут точно идентифицировать его, либо же склонны давать социально желательные ответы, продемонстрировать свою

[9] Marakshina et al. (2024)

[10] Since I conducted the survey in March, February was the latest month that I could refer to.

эмоциональную стойкость и подготовленность к предстоящему экзамену."

Therefore, in my main results (Table 6) I do not use PSS-10 as a proxy of stress. Instead, I use a natural experiment where a treatment group faced an additional stressor in the form of the canceled Olympiad, while a control group did not. By so, I create an exogenous variation in stress levels for two groups and get rid of the risk that some respondents may have reported unreliable or biased results. Certainly, my self-efficiency metrics may also be biased but I use them as controls instead of the main independent variable in the regression. I also use the PSS-10 in the IV model, so it's important to keep in mind that the scores may be underestimated, which could lead to underestimated estimates.

4.3 Variable Definitions

In this section I define main variables that I use in my regression analysis. Firstly, I specify stress metrics, secondly, I explain how my treatment and control groups are constructed. Finally, I define three types of academic success measurements that I use as dependent variables.

- **Covid PSS/ February PSS**: Perceived Stress Scales that refer to the Covid-19 lockdown and last February accordingly. They change in range between 0 and 40. The higher the PSS score is, the worse a respondent feels.

- **Covid/February Efficacy**: value of a perceived self-efficacy sub-scale. It ranges from 0 to 16. The higher the Efficacy score is, the less efficient a respondent tackles issues.

- **Covid/February Helplessness**: value of a perceived self-helplessness sub-scale. It ranges from 0 to 24. The higher the Helplessness score is, the more helpless a respondent feels.

- **Tried** is a dummy variable that is 1 if a participant has already took part in the Olympiad before, and zero if she participates for the first time. $Tried = 1$ defines my Control group.

- **D20=1** is an indicator of respondents from my Treatment group. My treatment group are school students who were subjected to the cancellation of the Olympiad in 2020.

- **Post=1** is an indicator of time periods that occurred after the Covid-19 lockdown. In the context of the All-Russian Olympiad and my research study, $Post = 1$ for years 2021, 2022 and 2023. I did not collect data from the 2024-participants, since they

were too young when Covid-19 was spreading out, thus they may have not been able to realize full consequences of Covid-19 pandemic and experienced low stress. I treat 2023-participants with caution as well. Age specific summary statistics proves my concern of that the younger a respondent was in 2020, the less stress they experienced.

- **Treat** is a variable that is equal to $Treat = D20 * Post$, and it deals with the issue that students from the Treatment group may have participated in the Olympiad before the pandemic exploded. If so, they are still students from the Treatment group but I should not include them in the Δ_{DiD} effect. In other words, $Treat$ is just a more compact designation of interaction between $D20$ and $Post$. It is extremely important to note that I treat the canceled Olympiad as an attempt, therefore $Treat = 1$ automatically means $Tried = 1$ as well.

- **R** is my main dependent variable, it stands for the respondent's academic achievements. As it was mentioned above, I use three different metrics of $R_{i,t}$ to allow for a robustness check. The first one is $Score_{i,t}$ that is $Score_{i,t} = \frac{(S_{i,t} - B_t)}{Total_t}$, where $S_{i,t}$ is a participant's score in year t, B_t is a boundary score for the participant's grade group[11], and $Total_t$ is a maximum possible score that a participants can gain in year t. Two other specifications of $R_{i,t}$ are $Percentile_{i,t}$ and $Win_{i,t}$. $Percentile_{i,t}$ is a respondent's percentile among her grade and year. Such metrics has better fit for grade and year-specific patterns (i.e. total amount of winners of the particular grade and particular year) that are orthogonal to the respondents' results.[12] $Win_{i,t}$ is a dummy variable that is equal to 1 if a respondent wins the Olympiad in year t, and zero if they participates but loses.

4.4 Summary Statistics

My data sample contains 149 unique respondents aged from 18 to 24, 53 of whom are girls and the rest 96 are boys. There are 44 respondents in my Treatment group, and there are 105 ones in the Control. An average respondent participates in the Olympiad roughly 1.5 times instead of one, therefore I have 232 unique participation-times. The distribution of observations is presented in the Table 3. I demonstrate structural differences between my self-collected sample and the official statistics in Table 3. My sample is skewed towards

[11] Boundary sco res are different for 9, 10 and 11 grades, and are unique for each Olympiad year.

[12] The number of winners in each category is decided by the judges through a voting process. Although the law requires at least 45% of all participants to be awarded, this percentage can differ significantly among grades. So, sometimes it is the 9_{th} grade-participants who receive more than 45% of winner places inside their grade, while sometimes the opposite happens and the 9_{th} grade gets less winners than the 11_{th} grade.

higher percent of female participants and more successful participants. I prove that this bias does not change the overall results in Section 9.

The main variables used in my regression analysis are presented in Table 1, where I test if there are any differences between treatment and control groups. One cannot report any statistically significant gap between treatment and controls means at any standard level of confidence.

Mean values of Covid and February PSS lay in the frames of a medium stress level following classification of Zolotareva (2023) who classifies PSS as low for PSS less than 13, medium for scores from 13 to 26, and high for scores from 27 to 40 that is a maximum possible one.

Table 1: T-test: Difference in Means between Treatment and Control

Variable	Treatment	Control	Difference	P-value
Covid PSS	17.36	18.56	-1.20	0.396
Covid Efficacy	6.95	7.41	-0.46	0.421
Covid Helplessness	10.41	11.15	-0.74	0.459
February PSS	18.00	18.19	-0.19	0.882
February Efficacy	6.80	6.61	0.19	0.730
February Helplessness	11.20	11.58	-0.38	0.630
Trials	1.52	1.57	-0.05	0.555
Age	21.00	20.75	0.25	0.509
Countryside	0.25	0.23	0.02	0.798
Wealth	0.20	0.06	0.14	0.329
Have a pet	0.36	0.46	-0.10	0.282

Note: P-values are from Welch's t-test.
"Have a pet" is a dummy for living with a pet during Covid-19 pandemics.
"Countryside" is a dummy for living in a countryside during Covid-19 pandemics.
Wealth is an index variable taking values from -2 to 2, where higher values stand for higher metrics, i.e. Wealth index equals to 2 means a respondent's family is the most wealthy, while -2 refers to the poorest one.

Table 2 compares Covid Stress and February Stress for different subsamples. I find that a null hypothesis of equal mean stress levels in 2020 and last February is rejected at 5% significance level for female part of the Control sample in favor of an alternative hypothesis about stress during quarantine being higher. However, the same is not true for boys from the Treatment sample, for whom we accept the null hypothesis. It does not lay in line with my initial assumption, yet I cannot guarantee that all response are trust-worthy. Turning to the Control sample, the null hypothesis is accepted for respondents aged 18 and higher

both for boys and girls at 10% level of confidence, as well as for girls aged 20 or higher. However, H_0 is rejected for boys when we drop eighteen and nineteen-years-old respondents from the Control sample. In other words, boys who are currently 18 or 19 years old reported statistically insignificant gap between Covid PSS score and February PSS score, probably because they were too young to estimate the scale of the Covid-19 pandemics in 2020.[13]

In the table 5 I demonstrate why I later control for sex in my regression analysis. Noting that my sample may not fully represent a real-Olympiad sample (see Table 3; I discuss it Section 9) and that I also have only 53 unique female respondents, I test difference in mean academic success by gender using official statistics.[14] The null hypothesis of equal means is rejected in 2018, 2019, and 2023 at 5% level of confidence for all three metrics of academic success in favor of that males get higher scores and win more frequently than females. However, a picture is different for 2021 and 2022 Olympiad years. In 2021 boys still won more often (p¡0.05), but there is no significant difference between boys and girls in relation to *Score* and *Percentile* at the same level of confidence. With P-values of 0.07 and 0.06 accordingly, gaps in *Score* and *Percentile* are only significant at 10% level of confidence. Finally, we accept H_0 of equal mean academic results for all 3 metrics for boys and girls in 2022 at 1%, 5%, and 10% levels of confidence.

To sum it all up, treatment and control samples do not vary between each other at any significant level of confidence. My sample does not ideally represent official population of the Olympiad participants but I show that it does not change main results in Section 9. An average respondent reports a higher stress level during Covid-19 lockdown relative to the last month stress but this gap is not always distinguishable from zero. The exception is boys from the Treatment group who reported the lowest Covid stress level among all other groups, though it contradicts common sense as well as other sub-samples statistics. According to the official Olympiad results, boys performed better than girls in 2018, 2019, 2021, and 2023 Olympiad years but the gap disappeared in 2022.

[13]Some respondents 18 or 19 years old told me that they were happy to stay home instead of attending school classes in person because it made it easier to skip classes and to mind their own business during lessons, i.e. chatting with friends online or playing computer games.

[14]I am thankful to Dmitry Dagaev and Danil Fedorovykh for providing me with access to this official statistics that was not published in the Net.

Table 2: T-test: Difference in Means between Covid PSS and February PSS

	N	Covid Mean	February Mean	Dif	s.e.	t-value	p-value
Male (1)	82	17.171	16.720	0.451	1.046	0.45	0.334
Female (1)	62	21.355	20.194	1.161	0.957	1.2	0.115
Male (2)	53	17.491	15.434	2.057	1.361	1.5	0.068
Female (2)	45	21.556	19.978	1.578	1.248	1.25	0.106
Male (3)	37	14.729	17.216	-2.486	1.673	-1.5	0.927
Female (3)	21	24	21.000	3	1.433	2.1	0.025

Note: $H_a : Dif > 0$

(1) - T-test by sex for all Control group.

(2) - T-test by sex for respondents from the Control group aged 20 or higher.

(3) - T-test by sex for Treatment.

Table 3: Distribution of the sample by subgroups.

	pre-Treatment	**post-Treatment**	**Girls**	**Boys**
D20=1	24	43	18	49
D20=0	37	128	62	103
Tried=1	10	96	35	71
Tried=0	51	75	45	81
Total	61	171	80	152

Note: observations are taken as participation, so that one respondent can be counted multiple times.

Table 4: Difference in Means between my sample and the official one.

Variable	Mean (Official)	Mean (My sample)	Difference	P-value
Female	0.234	0.330	0.096	0.006
Win	0.459	0.560	0.101	0.007
Percentile	50.523	55.745	5.222	0.011
Score	-3.162	-0.645	2.517	0.013
D20	0.219	0.306	0.087	0.011
Trials	1.484	1.464	-0.019	0.679

P-values from Welch's t-test.

Table 5: T-test: Difference in Mean Academic Results by Sex

Year	Variable	Mean (M)	Mean (W)	Difference	t-value	P-value
2018	Percentile	53.9	40.705	13.195	3.325	0.0012
2018	Score	-0.1	-6.3	6.2	2.824	0.0056
2018	Win	0.493	0.304	0.188	2.854	0.005
2019	Percentile	52.987	44.827	8.16	2.109	0.0367
2019	Score	-1.9	-6.8	4.9	2.384	0.0186
2019	Win	0.495	0.361	0.133	2.081	0.039
2021	Percentile	52.0	45.429	6.57	1.803	0.0735
2021	Score	-2.8	-6.7	4	1.899	0.0597
2021	Win	0.476	0.345	0.13	2.155	0.0328
2022	Percentile	50.678	50.998	-0.32	-0.085	0.9325
2022	Score	-2.3	-1.5	-0.7	-0.424	0.6722
2022	Win	0.477	0.475	0.001	0.018	0.9855
2023	Percentile	53.187	41.016	12.172	3.49	0.0006
2023	Score	-2.1	-7.3	5.2	3.001	0.0032
2023	Win	0.498	0.325	0.173	2.894	0.0044

Data source: official statistics. Welch's t-test was used; P-values are two-sided.

5 Identification strategy

5.1 Difference in Difference

My main underlying assumption is that those school students who were affected by the cancellation of the All-Russian Olympiad in 2020 faced an additional stressor during Covid-19 pandemic that the rest of the students who were not about to participate in the Olympiad in 2020 did not face.

Following this key assumption, I propose to use a Difference in Difference (DiD) approach. Unfortunately, I cannot test a parallel trend assumption for Treatment and Control groups in pre-Treatment periods simply because the Treatment group appears only in 2021 in the result of the cancellation of the Olympiad. Nevertheless, I employ the fact that some students from the Treatment group took part in the Olympiad before the Covid-19 pandemic expired, and include $D20$ dummy to show that the ex-ante students from the Treatment group did not outperform the average participant.

The main dependent variable is $R_{i,t}$ which is a proxy for academic achievements. Once again, I use three measures of it: first, $Score_{i,t}$ that is a percentage excess of the respondent's score obtained during the final stage of the Olympiad in a year t over a bounding score that determines which participants get the privilege and which do not. I divide this difference by the total score that could have been potentially obtained. It is important to use a reference-dependent measurement because of the fact that a passing grade differed from year to year, as well as a maximum number of points. Second, I use a binary variable $Win_{i,t}$ that equals to one if a participant won the Olympiad in a year t and is zero otherwise. Finally, I use $Percentile_{i,t}$ that is the participant's percentile inside her grade and year. The general plan is as follows:

1. Using my Control group to calculate by how much students who are participating repeatedly in year t are more likely to win the Olympiad this year; I call it $\Delta_{experience}$ and assume it to be above zero. Intuition is that there is an effect from scale in terms of preparation for the Olympiad, thus those students who participate repeatedly may have higher odds to win. It is true as such school students are more experienced and skilled. Variable that stands for the experience is $Tried_{i,t}$:

$$Tried_{i,t} = \begin{cases} 1, & \text{if a respondent has already participated in the Olympiad before} \\ 0, & \text{if a respondent is participating for the first time in year } t. \end{cases}$$

So, at the first step I calculate the following by-year value for for the Control group, assuming that this gap is persistent over time[15]:

$$\Delta_{control} = E[R_t|Tried = 1; D20 = 0] - E[R_t|Tried = 0]. \tag{1}$$

2. Next, I do the same for the Treatment group. To be more precise, I estimate a gap in academic results between students who participated for the first time and those who did it repeatedly and were objected to the cancellation of Olympiad in 2020[16]:

$$\Delta_{treatment} = E[R_t|Post = 1; D20 = 1] - E[R_t|Tried = 0]. \tag{2}$$

3. Finally, I compute $\Delta_{DiD} = \Delta_{treatment} - \Delta_{control}$ and attribute the obtained gap to the sum of a stress effect and a motivation effect.

The research design can be written in the form of a DiD regression model:

$$R_{i,t} = \alpha + \beta \cdot Tried_{i,t} + \gamma \cdot D20_i + \tau \cdot D20_i \cdot Post_t + A_{it} + \varepsilon_t \tag{3}$$

Where α is mean academic results for participants who take part in the Olympiad for the first time, β reports by how much school students do better when compete repeatedly, and τ is the desired effect of the shock. Negative τ stands for that a negative shock impaired academic success of school students, or in other words that a stress effect exceeded a positive motivation effect. In the further body of this paper I will examine the sign of τ by naming $\tau > 0$ as a positive DiD effect, and $\tau < 0$ as a negative DiD effect. ε_t is year Fixed Effects.

A_{it} in 3 is a list of control variables based mainly on Ye et al. (2019) results. They named motivation and psychological resilience as two main mediating factors that should be taken into account. In the frames of this study motivation is considered as a mechanism that runs parallel to the stress effect so I do not include it as a control variable. Psychological resilience includes personal traits such as competence, coping skills, self-efficacy, and environmental factors (parental support, community organization). I control for them with use of the PSS questionnaire mentioned above and with a set of personal questions[17].

[15] I show that allowing this gap to be different for post-treatment periods does not change anything much.

[16] Please note that condition on $Tried$ is not necessary for the first part of the equation 2, since the joint condition $D20 = 1; Post = 1$ automatically means that $Tried = 1$.

[17] See appendix for the full version of the questionnaire.

5.2 Instrumental variable approach

Since a motivation effect and an effect of stress are tightly intertwined with each other, DiD approach can only show which effect prevailed but it fails to estimate each effect separately. In this section I address the potential endogeneity caused by unobserved motivation effect by employing an instrumental variable approach. Specifically, I use a dummy "Having a pet"[18] as an instrument for Covid Stress for a male sub-sample. Indeed, there is a field of studies that prove pets to be an efficient tool to reduce stress (Beetz et al. (2012), Kertes et al. (2017)), and at the same time, pets are unlikely to affect motivation.

To prove that this IV approach is valid, I discuss the main identification assumptions that must hold in the next few paragraphs. First, the instrument I am using must be relevant. According to Figures 2 and 3 from the Appendix, "Dog dummy" variable has the strongest negative correlation with Covid PSS metrics for treatment boys (-0.24), while "Cat dummy" has the strongest correlation with Covid PSS for control boys. "Pet Dummy" is a more aggregated variable which allows for a greater variety of types of pets. I additionally prove that my instrument is valid by examining a first-stage regression results and F-statistics in Section 8.3.

Second, the instrument should satisfy the exclusion restriction, which means that having a pet should affect academic achievements exclusively through the stress reduction mechanism. Even if owning a pet may affect some adult's job career prospects through taking time in the morning and causing frequent delays, this may not be true of adolescents. It is parents who decide whether to have a new pet, and it is parents who generally take care of a pet, not a child. It is also highly unlikely that a pet could destroy the preparation process in any significant scale, since all participants took part in the Olympiad at will. Moreover, if a child won the Olympiad, it would be beneficial both for them and their parents, therefore the whole family would strive to eliminate negative consequences of a badly behaved pet. Besides all the discussed above, it is hard to come up with any other way how having a pet may affect academic achievements except from reducing distress levels, hence I claim that exclusion restriction is hold.

Third, the instrument must be exogenous. The main idea of the IV approach is to eliminate a motivation effect and to estimate a stress effect solely after controlling for the family wealth and size, region of living, etc. After doing so, a motivation effect should not be correlated with a pet ownership. Again, it is parents who decide whether to have a pet

[18] "Having a pet" is a dummy variable that equals to 1 if a respondent reported that she was living with a pet during the Covid-19 quarantine.

or not basing on the financial status of the family, their workload level, a number of children that need care and, finally, on their own perception to having a pet that is orthogonal to the child's motivation to win the Olympiad.

6 Results

In this section I describe the results of the estimations. Firstly, I provide the estimation results of my DiD regression 3 in order to determine which of the two effects prevailed in the Russian data. I want to recall that there is a large body of research that claims that high stress is the main barrier to academic success for young people (Lisnyj et al. (2021), Pascoe et al. (2020)). In the next few tables, I demonstrate that this may not be the case if one also takes a motivation effect into account. Then I allow for heterogeneity in the effects along stress-coping abilities, and try to isolate the stress effect from the motivation effect using an estimation of a 2SLS regression equation. Finally, I perform robustness checks on my results.

6.1 Main DiD results

The estimation results of the equation 3 are presented in Table 6. Since psychological research considers coping mechanisms to be different for boys and girls, I firstly estimate equation 3 for the whole sample controlling for the respondent's gender, and then I do the same for boys and girls separately. By so, columns 1-3 present the estimation results for the whole sample, while columns 4-6 do it solely for boys and columns 7-9 do — for girls. As it was already mentioned above, I use three different metrics for academic achievement, so in columns 1, 4, and 7 the dependent variable is $Score$. In columns 2, 5, and 8 it is $Percentile$, and in columns 3, 6, 9 it is Win. In my basic specification I control only for self-coping abilities that are presented by Low Efficacy dummy, since I have very limited number of observations to use as many controls as I wish to. In all presented estimation results Year Fixed Effects are absorbed, and standard errors are clustered by a county where a respondent spent the Covid-19 lockdown. The coefficient before $D20$ is whether

The Difference-in-Differences estimations in Table 6 show that historically those participants who participate repeatedly ($Tried_{i,t} = 1$) win in average by 17 p.p. ($p < 0.01$) more often, gain by 7.86 p.p. ($p < 0.05$) higher percentiles and score by 4.48 p.p. ($p < 0.05$) higher above the threshold than their peers who participate for the first time. Such results coincide with the general assumption that more experienced participants achieve higher scores due to

having had more years of preparation or being more familiar with the format.[19] Interestingly, I cannot reject the null hypothesis that male participants who participate for the second or third time perform as well as those who participate for the first time. This may be related to their overall attitude towards the Olympiad: it seems that once won, an average boy is less likely to continue tough preparation process compared to an average girl. It results in that one more trial is not associated with higher chance to win or to obtain higher scores for boys. Conversely, $Tried$ is a strong predictor of better outcomes for girls in all three scenarios.

According to columns 1 to 3, female students score on average 8.6 percentile points ($p < 0.05$) lower than boys and are awarded by 13% ($p < 0.05$) less frequently. This is a close approximation to the official statistics presented in the Table 5. Counterintuitively, male respondents reporting low stress-coping abilities (Low Efficacy) score higher on average and win more frequently compared to those with good coping skills. This contradicts my initial hypothesis. One possible explanation is that respondents with lower coping skills may also have had lower self-esteem, and as a result, they may underestimate their chances of winning, which leads to increased motivation for studying.

Switching to the main DiD coefficient before $D20 * Post$, it is positive and statistically significant at least at 5% for columns 1, 2, 3, 6, 7 and is significant only at 10% significance level for column 8. Overall, it means that on average, the motivation effect dominates the deteriorating stress effect, but this mechanism varies by gender. While girls from a treatment group have higher percentiles and higher scores relative to the control group, their probability of winning is unaffected. This means that the effect is strongest for those farthest from the boundary score, i.e., either the most skilled or least skilled participants. For those closer to the average, however, the effect is significantly weaker and there is no significant effect on winning probability. The mechanism is somewhat different for boys: the desired effect only occurs for column 6 (which corresponds to winning), while the influence on percentiles and scores is statistically zero.

[19] I do not include a dummy for that a respondent has already won the Olympiad before, since it correlates a lot with $Tried$, therefore, it would be a bad control.

Table 6: Main DiD results

	All			Boys			Girls		
	Score	Percentile	Win	Score	Percentile	Win	Score	Percentile	Win
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Tried	4.48**	7.86**	0.17***	2.45	4.13	0.10	8.38***	14.89**	0.37*
	(1.70)	(3.41)	(0.06)	(2.05)	(4.81)	(0.10)	(1.38)	(4.88)	(0.19)
D20	-9.60**	-16.34**	-0.45***	-5.86	-8.77	-0.48***	-14.56*	-27.49*	-0.31
	(3.85)	(6.61)	(0.09)	(5.92)	(8.15)	(0.11)	(7.08)	(14.28)	(0.24)
D20*Post	8.68**	17.94**	0.44***	4.20	8.20	0.49***	16.61**	36.12*	0.24
	(4.16)	(7.54)	(0.15)	(6.01)	(8.76)	(0.16)	(7.44)	(17.08)	(0.32)
Girls	-3.77*	-8.61**	-0.13**						
	(2.08)	(4.08)	(0.06)						
Low Eff.	6.42***	12.11***	0.08**	8.75***	14.24***	0.12*	2.07	5.93	0.02
	(1.20)	(2.44)	(0.03)	(1.40)	(3.49)	(0.06)	(2.19)	(3.73)	(0.08)
Constant	-0.92	54.84***	0.57***	-0.79	55.48***	0.59***	-5.05	44.97***	0.38***
	(1.66)	(2.60)	(0.04)	(1.76)	(3.01)	(0.06)	(3.00)	(4.81)	(0.08)
N	232	232	232	152	152	152	80	80	80
FE	Year	Year	Year	Year	Year	Year	Year	Year	Year
Clustered s.e.	Region	Region	Region	Region	Region	Region	Region	Region	Region

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Low Efficacy is a dummy variable that equals to one if a respondent reported Covid PSS score higher than 27 out of 40.

6.2 Heterogeneity Check

To examine whether the motivation and stress effects vary by stress-resilience, Table 7 augments the DiD model with an interaction between a post-treatment group and low-efficacy coping skills dummy. Thus, my main equation 3 becomes the following:

$$\begin{aligned} R_{i,t} &= \alpha + \beta_1 \cdot Tried_{i,t} + \beta_2 \cdot Low\ Efficacy + \beta_3 \cdot D20_i + \beta_4 \cdot D20_i \cdot Post_t \\ &\quad + \tau \cdot D20_i \cdot Low\ Efficacy \cdot Post_t + A_{it} + \varepsilon_t \\ &= \alpha + \beta_1 \cdot Tried_{i,t} + \beta_2 \cdot Low\ Efficacy + \beta_3 \cdot D20_i + \beta_4 \cdot Treat_{i,t} \\ &\quad + \tau \cdot Low\ Efficacy \cdot Treat_{i,t} + A_{it} + \varepsilon_t \end{aligned} \tag{4}$$

After allowing for heterogeneous effects along stress-coping abilities, the results change a lot. The results in columns 1-3 of Table 7, which refer to the whole sample, suggest that the magnitude of the treatment effect on scores and percentiles is significantly higher for participants who are accustomed to stress compared to those who are stress-resistant (columns 1 and 2). However, there is no significant difference in the probability of winning for the two types of participants in column 3, as both low-efficient and stress-resistant respondents score 41 percentage points better than the control group.

Next, I compare heterogeneity of the effect for boys and girls separately. Columns 7 to 9 in Table 7 claim that a positive success premium remains the same for the least and for the most stress-resistant girls. Moreover, its average magnitude has remained relatively compared to Table 6 (Columns 7-9). On the contrary, results on male sub-sample turn out to be highly heterogeneous along stress coping abilities: while there is no significant difference between the control and treatment groups in columns 4 and 5 of Table 6, the main DiD effect among low-efficient adolescents in Table 7 becomes highly positive and significant ($p < 0.01$). The effect on stress-resistant boys in the Treatment group remains negligible. These findings suggest that, while girls' motivational boost is fairly consistent, boys' gains are concentrated among those who are already accustomed to stress. Stress-resistant boys, on the other hand, report no treatment effect whatsoever.

6.3 IV results

Table 8 reports two-stage least-squares estimates that takes Covid PSS score as endogenous and use living with a pet during the 2020 lockdown as an instrument to obtain causal effects of stress. I control for family wealth, size, county of living during Covid-19 lockdown and for time of participation in the Olympiad. I do so in order to address endogeneity problem

discussed in Section 6.2. Indeed, financial well-being, the number of children in the family, and the region of residence are the main factors that determine a parent's willingness to own a pet, therefore I include them in the estimated equation. Notably, I present IV results for boys only because of the fact that having a pet turned out to be an invalid instrument for girls, moreover correlation between girls' stress level and presence of a pet turned out to be positive, which contradicts common sense. Therefore, I do not present 2SLS estimation results for girls.

For boys, the first-stage coefficient on the pet dummy (Column 4) after controlling for family wealth, size, year and region of living, is -0.77 ($p < 0.01$), a Kleibergen-Paap F-statistic is 15.8, which is high above the Stock–Yogo 10 % weak-instrument critical value. This implies that the chosen instrument is strong enough to partial out endogenous part of variation of stress.

Switching to the 2SLS second stage estimations (Columns 1-3), none of the stress coefficients is statistically different from zero, though all of them are negative. By so, once motivation is purged via the instrument variable, stress per se does not effect Olympiad performance of boys anyhow.[20]

[20] However, the sign of the stress effect is negative, which may refer to that I have not enough observations. Further research is needed to state it with certainty.

Table 7: DiD results with heterogeneous effects

	All			Boys			Girls		
	Score	Percentile	Win	Score	Percentile	Win	Score	Percentile	Win
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Tried	4.83***	8.48**	0.18***	2.87	4.66	0.10	8.46***	15.21***	0.37*
	(1.64)	(3.41)	(0.06)	(1.95)	(4.75)	(0.10)	(1.39)	(4.80)	(0.20)
D20	-9.90**	-16.86**	-0.45***	-5.68	-8.55	-0.48***	-14.62*	-27.73*	-0.31
	(3.95)	(6.87)	(0.09)	(5.98)	(8.31)	(0.11)	(7.11)	(14.33)	(0.24)
Low Eff.	4.79***	9.30***	0.06	6.19**	11.04**	0.08	1.83	4.95	0.02
	(1.60)	(2.95)	(0.05)	(2.34)	(4.67)	(0.08)	(2.43)	(4.13)	(0.08)
Treat	7.00	15.03*	0.41**	1.81	5.21	0.45**	16.18*	34.35*	0.24
	(4.24)	(8.01)	(0.17)	(6.19)	(9.69)	(0.17)	(7.73)	(17.67)	(0.35)
Low Eff.*Treat	10.00**	17.26***	0.14	24.41***	30.57***	0.34**	1.17	4.80	-0.01
	(4.85)	(6.21)	(0.17)	(3.71)	(7.57)	(0.15)	(2.02)	(4.97)	(0.11)
Girls	-4.16*	-9.29**	-0.14**						
	(2.20)	(4.37)	(0.06)						
Constant	-0.55	55.48***	0.57***	-0.54	55.79***	0.59***	-5.01	45.15***	0.38***
	(1.72)	(2.64)	(0.04)	(1.85)	(3.12)	(0.07)	(3.06)	(4.95)	(0.08)
N	232	232	232	152	152	152	80	80	80
FE	Year	Year	Year	Year	Year	Year	Year	Year	Year
Clustered s.e.	Region	Region	Region	Region	Region	Region	Region	Region	Region

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Low Eff. is "Low Efficacy": it is a dummy variable that equals to one if a respondent reported Covid PSS score higher than 27 out of 40.
Treat=D20*Post.

Table 8: 2SLS estimations results

(a) Boys

	2nd Stage			1st Stage
Variable	Score	Percentile	Win	Covid Stress
	(1)	(2)	(3)	(4)
Covid Stress	-0.25	-4.66	-0.13	
	(2.83)	(6.40)	(0.11)	
Pet Dummy				-0.77***
				(0.19)
Tried	2.20	6.85	0.17	0.38**
	(2.60)	(5.39)	(0.11)	(0.16)
Wealth	-1.00	-3.72	-0.06	-0.23**
	(1.22)	(2.44)	(0.04)	(0.10)
Family size	-1.21*	-2.85*	-0.05*	-0.01
	(0.73)	(1.46)	(0.03)	(0.09)
Constant	7.13	85.94***	1.12***	3.39***
	(7.64)	(19.88)	(0.29)	(0.45)
First-stage F	15.78	15.78	15.78	
N	152	152	152	152

Instrumental variable for boys is Pet Dummy: dummy for living with a pet during Covid-19 quarantine. Standard errors clustered by region of living in parentheses. Year fixed effects absorbed. Kleibergen-Paap Wald rk F statistic included. Stock-Yogo weak ID test critical values: 10% maximal IV size 16.38. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Robustness Checks

To increase the credibility of the results, I conduct some robustness tests for my DID approach. In addition to using three measures of academic achievement throughout the paper, I add a set of controls to the main DiD specification, allowing for time varying means in the Control group, and re-run regressions for the main DiD specification 3 with a full set of controls, weighting my observations in a way to eliminate selection bias from the collected data set.

To begin with, I control for Family Wealth and Family Size in Table 9, to show that the main results do not change much. I choose them as the most credible controls because

they can be considered as proxies for motivation to win. Indeed, Family Wealth turns out to be negatively correlated with girls' academic achievements in columns 7-9, and one possible explanation is that students from more wealthy families have lower incentives to prepare so seriously, since they know that they will be able to enroll in a prestigious university on a fee-based basis. The same is true of Family Size which might be correlated with wealth as well as with the how comfortable it is to be at home.

According to Table 9, the main DiD results stay similar in all alternative specifications but their magnitude slightly decreases (from 8.68 in Table 6 Column 1 to 8.12; from 17.94 in in Table 6 Column 2 to 16.82). The standard errors of the main effect in Columns 7 to 9 increase, which implies that the motivation effect contribution to the total DiD effect was partialed out by the added controls.

Additionally, I have included interaction between $Tried_{i,t}$ and $Post$ in my main specification, presented in Section 5.1, to allow for a different time trend in the Control group during post-Covid periods compared to earlier ones. Looking at Columns 1-3 of Table 10, participants from the Control group who participate repeatedly after 2020 gain on average 15.24 percentage points (p.p.) lower scores, perform 30.33 percentage points worse than others, and have a 0.52 percentage point lower probability of winning than respondents from the same group prior to the Covid-19 lockdown.

Summing up the coefficients before $Tried_{i,t}$ and $Post$, we can see that participants from the Control group, that is, those who participate repeatedly but were not affected by the cancellation of the Olympiad in 2020, performed in years 2021, 2022, and 2023 almost as well as those who participated for the first time.[21].

However, such a change does not influence the main DiD results: the desired effect stays positive and statistically significant for Columns 1, 2, 3, 6, 7, and 8 but its size slightly varies: that is, the DiD effect becomes higher for boys (from 0.45 to 0.56 for probability to win in Column 6) and drops in size for girls (from 16.61 to 15.70 for scores in Column 7).

Finally, I re-weight my observations to get rid of the bias towards more experienced participants in my sample (see Table 4). To do so, I attribute relatively lower weights to winners compared to the non-winners. I take $w_{winner} = \frac{0.459}{0.560} = 0.819$, while keeping $w_{non-winners} = 1$. The goal of this re-balance is to approximate the average values of my sample to the official one. Indeed, 0.459 is a fraction of winners in the full population, whereas the same value in my self-collected data is 0.560. The proposed weighting leads to

[21]For example, by summing the corresponding coefficients in Column 1, I see that the average participant in the Control group scored only 0.55 percentage points more than first-timers and had only 4 percentage points higher odds of winning.

exactly the same proportion of winners in all participants. The results after the weighting procedure are presented in Table 11. The DiD effect drops in size and becomes statistically insignificant for all three dependent variables for girls (Columns 7 to 9) relative to the same columns in Table 9 but the effect remains the same (it equals to 0.48, $p < 0.01$) for probability to win for boys (Column 6). This implies that the obtained results for females are more credible for the most successful girls, while the effect on boys is uniform and unbiased.

All in all, the signs of the effects stay positive and close in size to the ones in the baseline, but sometimes the magnitude decreases and makes the coefficients insignificant. One still cannot report any non-zero effects for boys' scores and percentiles, though the effect on the probability to win is strongly positive ($p < 0.01$) and large in size (around 50 p.p. raise in probability to win) in all the presented specifications. This brings an additional confirm to that the main effect for males is referred to the boys who scored points close to the boundary score, but the effect for the most and for the least competitive boys is close to zero. Conversely, the effect for girls is present only for the higher scores but not for the higher probability to win. This suggests that the effect is more pronounced for those who are further from the boundary score, that is, for those for whom a 36 percentage point increase in percentile does not change the outcome of the Olympiad. The DiD effects for girls become statistically insignificant after the weighting procedure discussed above, but they remain of the same magnitude.

7 Conclusion

To summarize, this paper examines the two-fold impact of negative shocks on the academic achievement of school students in settings where the stakes are high. I hypothesize that in the result of a negative shock students become more exhausted due to the higher stress as well as more motivated to study hard, since the odds to succeed decrease. I exploit a unique case of cancellation of a national stage of the All-Russian Olympiad in economics in 2020 to create exogenous variation in both long-lasting stress and perceived stakes among otherwise comparable cohorts of Russian adolescents. I firstly use the Difference-in-Difference approach to determine how well students from the Treatment group performed relative to their peers from the Control group. Next, I switch to the 2SLS regression design to isolate the deteriorating effect from a motivational one. The study provides the following insights.

1. Motivation can outweigh the psychological burden of stress. Difference-in-Difference estimations and multiple robustness checks show that the cohort exposed to the

2020 cancellation out-performed a comparable group of students who participated in the Olympiad repeatedly but were not exposed to the negative shock. In the conceptual framework of this study it means that the "motivation effect" substantially exceeded the "stress effect", when stakes got higher.

In other words, when an adverse shock simultaneously raises the perceived value of success, motivated students—especially those accustomed to coping with uncertainty—may transform anxiety into additional effort.

2. The net effect is heterogeneous along stress-coping abilities. Adding interactions between treatment status and stress-coping abilities revealed the effects to be heterogeneous. Among boys, it is only probability to win that the shock has influenced on. Higher odds to succeed are fully attributed to the least resistant male participants, probably, because they underestimated their odds to win and studied more. Since there is no premium for scores and percentiles, I suggest that the attributed effect is the strongest for the medium-skilled boys, who jump over the boundary score and become winners. For stress-resistant males, I cannot report any statistically significant differences between them and the Control sample. Among girls, the premium seems to appear in scores and percentiles, regardless of the type of stress resistance, but not in higher odds to win. This implies that motivation chiefly benefited girls already far from the cut-off.

3. Stress per se carries a close-to-zero negative impact. Instrumental-variable approach enables me to partial out the upward bias from motivation and show small, statistically insignificant negative coefficients on the Covid Stress for boys. When combined with the DiD results, this suggests that stress alone does not significantly hinder Olympiad performance unless it is not accompanied by a corresponding increase in perceived importance of the Olympiad.

It should also be noted that the considered question is not fully answered and requires further research. First, I conduct my research using small and skewed towards high achievers data sample, which raise a doubt according to the internal validity of the results. The results of the weighting procedure have only enhanced such concerns. Self-reported PSS scores are highly likely to be understated, especially for boys. Though I was able to get around this problem by using exogenous variation in stress in my DiD approach, 2SLS estimation results may still be underestimated. Finally, I lacked observations to investigate differences between the mechanisms for 10th and 11th-graders. Further steps may complement my analysis as well as highlight the need in tailored educational approach for stress-resistant students and their peers susceptible to stress.

Table 9: DiD results: Robustness Check

	All			Boys			Girls		
	Score	Percentile	Win	Score	Percentile	Win	Score	Percentile	Win
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Tried	4.71**	8.32**	0.18***	2.86	5.04	0.11	7.61***	13.74**	0.36
	(1.80)	(3.43)	(0.06)	(2.18)	(5.11)	(0.10)	(1.47)	(5.87)	(0.21)
D20	-9.07**	-15.27**	-0.44***	-5.91	-8.04	-0.49***	-14.58*	-27.44*	-0.33
	(3.79)	(6.58)	(0.10)	(5.68)	(7.69)	(0.11)	(7.27)	(14.92)	(0.26)
D20*Post	8.12**	16.82**	0.42***	3.74	6.77	0.48***	16.29*	35.61	0.24
	(3.98)	(7.53)	(0.15)	(5.69)	(8.40)	(0.16)	(9.05)	(20.30)	(0.39)
Girls	-4.00*	-9.07**	-0.15**						
	(2.18)	(4.04)	(0.06)						
Low Eff.	6.23***	11.72***	0.08**	8.57***	13.63***	0.11	1.13	4.50	0.01
	(1.30)	(2.51)	(0.04)	(1.41)	(3.42)	(0.07)	(1.55)	(3.23)	(0.09)
Wealth	-1.24	-2.49	-0.03	-0.26	-1.47	0.00	-3.25**	-4.77**	-0.07***
	(0.82)	(1.57)	(0.03)	(0.98)	(2.14)	(0.04)	(1.15)	(1.69)	(0.02)
Family size	-0.46	-0.91	-0.03	-1.25	-2.49	-0.04	1.16	1.79	0.00
	(0.64)	(1.00)	(0.02)	(0.89)	(1.72)	(0.03)	(0.95)	(1.14)	(0.02)
Constant	0.48	57.64***	0.65***	2.87	62.78***	0.72***	-7.31***	41.44***	0.38***
	(2.67)	(4.69)	(0.08)	(4.05)	(7.04)	(0.13)	(1.32)	(3.54)	(0.09)
N	232	232	232	152	152	152	80	80	80
Sample	All	All	All	Boys	Boys	Boys	Girls	Girls	Girls
FE	Year	Year	Year	Year	Year	Year	Year	Year	Year
Clustered s.e.	Region	Region	Region	Region	Region	Region	Region	Region	Region

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Low Eff. is "Low Efficacy": it is a dummy variable that equals to one if a respondent reported Covid PSS score higher than 27 out of 40. Wealth is an index variable. Countryside is a dummy for that a respondent lived in a countryside during quarantine.

Table 10: DiD results with post*tried

	All			Boys			Girls		
	Score	Percentile	Win	Score	Percentile	Win	Score	Percentile	Win
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Tried	15.79***	30.38***	0.56***	15.32***	29.22***	0.60***	14.72**	27.04**	0.60***
	(4.64)	(6.56)	(0.07)	(4.99)	(7.56)	(0.13)	(6.35)	(10.48)	(0.10)
Tried*Post	-15.24***	-30.33***	-0.52***	-16.41***	-32.00***	-0.64***	-9.27	-17.77	-0.34
	(5.51)	(9.20)	(0.13)	(5.86)	(9.68)	(0.16)	(8.88)	(19.04)	(0.32)
D20	-7.20	-11.55*	-0.37***	-4.99	-7.07	-0.44***	-11.31	-21.26	-0.19
	(4.34)	(6.43)	(0.10)	(6.34)	(8.65)	(0.13)	(8.11)	(16.40)	(0.28)
D20*Post	9.42**	19.41**	0.46**	6.14	11.99	0.56***	15.70*	34.37*	0.20
	(4.20)	(7.24)	(0.17)	(6.22)	(9.88)	(0.20)	(7.40)	(16.89)	(0.32)
Girls	-3.69*	-8.46**	-0.13**						
	(2.06)	(4.02)	(0.06)						
Low Eff.	6.36***	11.98***	0.08**	8.64***	14.03***	0.11*	2.28	6.33	0.03
	(1.05)	(1.87)	(0.03)	(1.20)	(2.64)	(0.06)	(2.08)	(3.58)	(0.07)
Constant	-0.63	55.42***	0.58***	-0.32	56.38***	0.61***	-4.99	45.09***	0.38***
	(1.41)	(2.27)	(0.04)	(1.53)	(2.74)	(0.06)	(2.81)	(4.49)	(0.08)
N	232	232	232	152	152	152	80	80	80

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Low Eff. is "Low Efficacy": it is a dummy variable that equals to one if a respondent reported Covid PSS score higher than 27 out of 40. Wealth is an index variable. Countryside is a dummy for that a respondent lived in a countryside during quarantine.

Table 11: DiD results: Weighting

	All			Boys			Girls		
	Score	Percentile	Win	Score	Percentile	Win	Score	Percentile	Win
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Tried	4.89**	8.62**	0.18***	2.88	4.99	0.11	8.22***	14.88**	0.37
	(1.79)	(3.54)	(0.07)	(2.23)	(5.26)	(0.11)	(1.46)	(6.03)	(0.22)
D20	-8.43**	-13.99**	-0.44***	-5.36	-7.06	-0.49***	-13.80*	-26.02*	-0.32
	(4.00)	(6.60)	(0.10)	(6.09)	(7.84)	(0.11)	(6.95)	(14.09)	(0.24)
D20*Post	7.20*	15.08*	0.41**	2.88	5.52	0.48***	14.98	33.05	0.21
	(4.14)	(7.59)	(0.16)	(6.07)	(8.52)	(0.16)	(8.81)	(19.90)	(0.38)
Girls	-3.97*	-9.02**	-0.15**						
	(2.18)	(4.03)	(0.06)						
Low Eff.	6.39***	12.30***	0.09**	8.62***	14.07***	0.12	1.40	5.09	0.01
	(1.32)	(2.51)	(0.04)	(1.50)	(3.58)	(0.07)	(1.57)	(3.24)	(0.09)
Wealth	-1.34	-2.57	-0.03	-0.30	-1.47	0.00	-3.34**	-4.82**	-0.07***
	(0.84)	(1.61)	(0.03)	(1.02)	(2.20)	(0.04)	(1.18)	(1.73)	(0.02)
Family size	-0.52	-1.01	-0.03	-1.35	-2.67	-0.04	1.10	1.69	0.00
	(0.64)	(1.01)	(0.02)	(0.94)	(1.78)	(0.03)	(0.91)	(1.08)	(0.02)
Constant	-0.36	55.63***	0.60***	2.25	61.12***	0.67***	-8.26***	39.31***	0.34***
	(2.82)	(4.83)	(0.08)	(4.35)	(7.34)	(0.13)	(1.28)	(3.44)	(0.09)
N	232	232	232	152	152	152	80	80	80
Sample	All	All	All	Boys	Boys	Boys	Girls	Girls	Girls
FE	Year	Year	Year	Year	Year	Year	Year	Year	Year
Clustered s.e.	Region	Region	Region	Region	Region	Region	Region	Region	Region

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Low Eff. is "Low Efficacy": it is a dummy variable that equals to one if a respondent reported Covid PSS score higher than 27 out of 40.

Wealth is an index variable. Countryside is a dummy for that a respondent lived in a countryside during quarantine.

Observations are weighted. For winners w=0.8196, for non-winners it is 1.

8 Appendix

8.1 Theoretical Model

I explore the research question in greater detail using a modified model by Wälde (2018). Although the authors consider a wide range of issues, such as types of stressors that cause stress, coping mechanisms that have different efficiency, and personal traits that make differences in relation to influence of stress on income, I will generally base on the 5th section, named "Stress, productivity and income". So, let me consider a simple model that will clearly demonstrate the current question of this paper.

The starting point is definitions of the key concepts that I am using below.

- Psychological stressors are "social and physical environmental circumstances that challenge the adaptive capabilities and resources of an organism"[22]; appearance of stressors is the first step of stress occurrence. Stressors cause a change of stress level.

- Appraisal is "the mechanism through which stressors increase emotional tension"[23]. Generally speaking, appraisal process is responsible for how much stress level is changed because of one particular stressor. Appraisal processes vary a lot depending on personal characteristics. For instance, the same stressors can cause different stress surges due to different appraisal mechanisms.

- Working memory, following Smith and Kosslyn (2007), is "the system that enables the short-term storage and mental manipulation of information". The next definition is taken from the same source.

- Working memory capacity (working memory span) is "a measure that describes the maximum amount of information an individual can store in working memory".

In other words, every human is endowed with a certain amount of working memory M[24] that is essential for any cognitive actions, from choosing what to eat and what to put on to solving intellectual problems at university or at work. In my model that is a slight modification of Wälde (2018), working memory is essential for preparation to the All-Russian Olympiad. From other side, working memory takes part in an appraisal process:

[22]Definition is taken from Monroe and Slavich (2016)
[23]From Wälde (2018)
[24]Designations are taken from Wälde (2018)

"the appraisal process evoked by stressors is one such example of an automatic, exogenous, allocation of working memory resources. As stressors are exogenous to the individual, the amount of working memory used for the appraisal process is exogenous as well", - Wälde (2018) says.

Now we are ready to build a model. Let p be a stressor, $W_t(p)$ be a function of an appraisal process at certain time point. Then the following equation takes place, since working memory is spent on coping with stress and preparation to the Olympiad only. M is set exogenously and is the same for all individuals in the model.

$$M = M(W(p,t)) + M(a(t)) \tag{5}$$

Indeed, $M(W(p,t))$ is the amount of working memory, that is spent on the appraisal process, that is, how much cognitive energy is needed to experience stress, that was caused by a stressor p. As well as the authors do, I demand that a higher stress level requires more working memory to handle it, so $M'(W(p,t) > 0$. The rest of the working memory can be used for everyday activities, but in my model, it is used to pay attention $a(t)$ to the preparation for the Olympiad. For simplicity I assume $M(a(t)) = a(t)$ as well as $M(W(t)) = kW(t)$. If so, 5 converges into

$$M = kW(t) + a(t)) \tag{6}$$

The main dependent variable in the model is $R(t)$, it is so-called readiness for the Olympiad. It is a cumulative measure of how well a child is prepared for the Olympiad, taking into account her initial ingenuity, mathematical skills, and intuition, marked as h. The readiness $R(t)$ can also be interpreted as the odds of winning the Olympiad at moment t. $R(t)$ is also determined by labor amount $l(a(t))$, it is the amount of hours spent on the preparation, and by effort e. All the discussed above sum in the following equation:

$$R(t) = l(a(t))eh = [M - k \cdot W(t)]eh. \tag{7}$$

As $l(a(t))$ rises in attention $a(t)$, the odds of winning rises in attention as well. As by 6 attention falls in stress, readiness falls in stress too. It is intuitively clear: attention to studies measures how easy an individual finds it to concentrate on preparation process. If it is easy, she studies more, so that l raises. Otherwise, if it is hard to focus, one would study less. How easy it is to focus on studies is a reverse function of stress, so the odds of winning decrease in stress as well.

However, such a simple model cannot explain the positive impact of academic stress,

i.e. competitive atmosphere or tight deadlines, therefore I align with Wälde (2018) by differentiating between two types of stressors.

Stressors on studies, i.e. highly competitive atmosphere or demanding parents, induce effort e as a response, and as an individual strives to minimize effort by nature, I assume that effort that she chooses is equal to study stressors. Recalling 7, Readiness R increases in effort, thus following Wälde (2018) I will name such stressors productive. Private stressors do not have any productive nature in average, so they are called p^{pure}. It results in

$$p = e + p^{pure} \tag{8}$$

By substituting 8 into 7 we obtain:

$$R(t) = l(a(t))eh = [M - k \cdot W_t(p)]eh = [M - k \cdot W_t(e + p^{pure})]eh. \tag{9}$$

While effort has a direct positive effect on Readiness by increasing the efficiency of studies, higher effort is also a more severe stressor, and, sequently, leads to higher stress W_t that is a reason of attention's deterioration. By so, study stressors present a trade-off between a motivation to put higher effort to studies and limited Working Memory that is available for studies: the more is used on stress, the less remains for studies.

8.2 Questionarie

- Пожалуйста, отметьте, что вам есть 18 лет, и вы готовы пройти опрос.
- Выберите Ваш пол.
  М; Ж
- Сколько Вам лет?
- Сколько раз вы участвовали в ЗЭ ВСОШ по экономике? Пожалуйста, не считайте отмененную олимпиаду 2020 года за участие.
  Никогда не участвовал(а); 1; 2; 3 или более
- Вспомним, что все заключительные этапы ВСОШ были отменены в 2020 году в связи с пандемией Covid-19. Вы должны были участвовать в ЗЭ ВСОШ по экономике в 2020 году?
  Да; Нет

Далее респонденты указывали свой балл на Заключительном этапе во все года, в которые они участвовали в ЗЭ ВСОШ по экономике. Респондентам предоставлялась ссылка на Google-таблицу с протоколом заключительного этапа олимпиады за тот год, в который респондент участвовал в олимпиаде. Далее респонденту нужно было найти себя в списке участников, а также свой итоговый балл за олимпиаду и указать его в качестве ответа на вопрос. Таким образом я смогла сделать опрос анонимным. Ниже приведен пример формулировки вопроса.

- Пожалуйста, напишите ваш балл на заключительном этапе ВСОШ по экономике в 2023 году.

  Таблицы с результатами находятся здесь: Протокол ЗЭ ВСОШ по экономике - 2023. Все просто: найдите себя в списке участников, посмотрите ваш балл и напишите его ниже!

- За какой класс вы выступали в 2023 году?
  9 класс; 10 класс; 11 класс

- В каком регионе России Вы жили во время режима самоизоляции, введенного для борьбы с пандемией Covid-19?

- Выберете количество человек, которые проживали с Вами в одной квартире/доме во время режима самоизоляции (НЕ включая Вас самих).
  0; 1; 2; 3; 4; 5 или более

- Выберите все верные для Вас утверждения. Во время режима самоизоляции 2020 года:
  Я делил(а) свою комнату с кем-то еще.
  Я жил(а) в одной квартире/доме с маленьким ребенком.
  Я занимался дополнительной физической активностью (спортом).
  Ничего из вышеперечисленного.

- Был ли у вас домашний питомец во время режима самоизоляции в 2020 году?
  Нет; Да, собака; Да, кошка; Другое

- Во время режима самоизоляции Вы жили в городе или за городом?
  Жил(а) в городе; Жил(а) за городом — на даче; Другое

- Сколько раз в неделю в среднем Вы выходили из дома во время режима самоизоляции в 2020 году? Считайте любые выходы на улицу: выбросить мусор, погулять с собакой, сходить в магазин и т.д. Если Вы жили на даче, считайте выход на территорию участка также выходом на улицу.
  Вообще не выходил(а)
  Выходил(а) 1-2 раза в неделю
  Выходил(а) 3-5 раз в неделю
  Выходил(а) практически ежедневно
  Выходил(а) практически ежедневно, иногда — по несколько раз в день
  Другое

- Как бы Вы могли описать материальное положение вашей семьи во время режима самоизоляции 2020 года?
  Денег не хватало даже на питание.
  На питание денег хватало, но покупка одежды вызывала серьезные проблемы.
  Денег хватало на питание и одежду, но купить телевизор, холодильник или стиральную машину было трудно.
  Наших заработков хватало на все, кроме таких дорогих приобретений, как дача, квартира.
  Материальных затруднений не испытывали. При необходимости могли приобрести дачу, квартиру.

- Пожалуйста, оцените Ваш общий уровень стресса за последний месяц от 1 до 5, где 1— стресс полностью отсутствует, 5 — постоянно нахожусь в состоянии стресса.
  1; 2; 3; 4; 5

- Пожалуйста, оцените Ваш средний уровень стресса во время режима самоизоляции в 2020 году, где 1 — стресс полностью отсутствует, 5 — постоянно нахожусь в состоянии стресса.
  1; 2; 3; 4; 5

Прочитайте, пожалуйста, следующие утверждения и оцените, как часто **за последний месяц**:

1. Вы расстраивались из-за того, что происходило неожиданно?

2. Вы чувствовали, что не способны контролировать важные моменты своей жизни?

3. Вы нервничали и испытывали стресс?
4. Вы чувствовали уверенность в своей способности справляться с личными проблемами?
5. Вы чувствовали, что все идет так, как это нужно Вам?
6. Вы обнаруживали, что не можете справиться со всем, что Вам приходилось делать?
7. Вы могли контролировать свою раздражительность?
8. Вы чувствовали, что находитесь на вершине успеха?
9. Вы злились из-за того, что происходило без контроля с Вашей стороны?
10. Вы чувствовали, что Вам не преодолеть всех скопившихся трудностей?

0 — Никогда
1 — Почти никогда
2 — Иногда
3 — Часто
4 — Очень часто

Прочитайте, пожалуйста, следующие утверждения и оцените, как часто **в период самоизоляции 2020 года**::

1. Вы расстраивались из-за того, что происходило неожиданно?
2. Вы чувствовали, что не способны контролировать важные моменты своей жизни?
3. Вы нервничали и испытывали стресс?
4. Вы чувствовали уверенность в своей способности справляться с личными проблемами?
5. Вы чувствовали, что все идет так, как это нужно Вам?
6. Вы обнаруживали, что не можете справиться со всем, что Вам приходилось делать?
7. Вы могли контролировать свою раздражительность?

8. Вы чувствовали, что находитесь на вершине успеха?

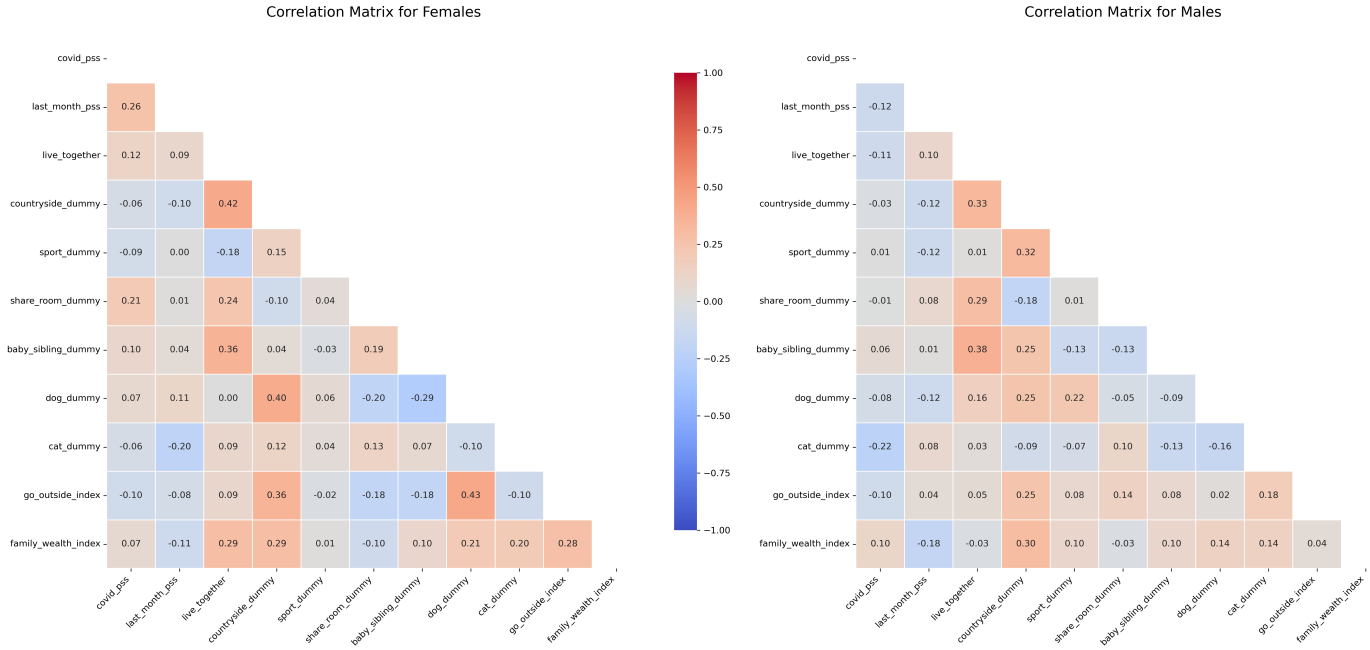
9. Вы злились из-за того, что происходило без контроля с Вашей стороны?

10. Вы чувствовали, что Вам не преодолеть всех скопившихся трудностей?

0 — Никогда
1 — Почти никогда
2 — Иногда
3 — Часто
4 — Очень часто

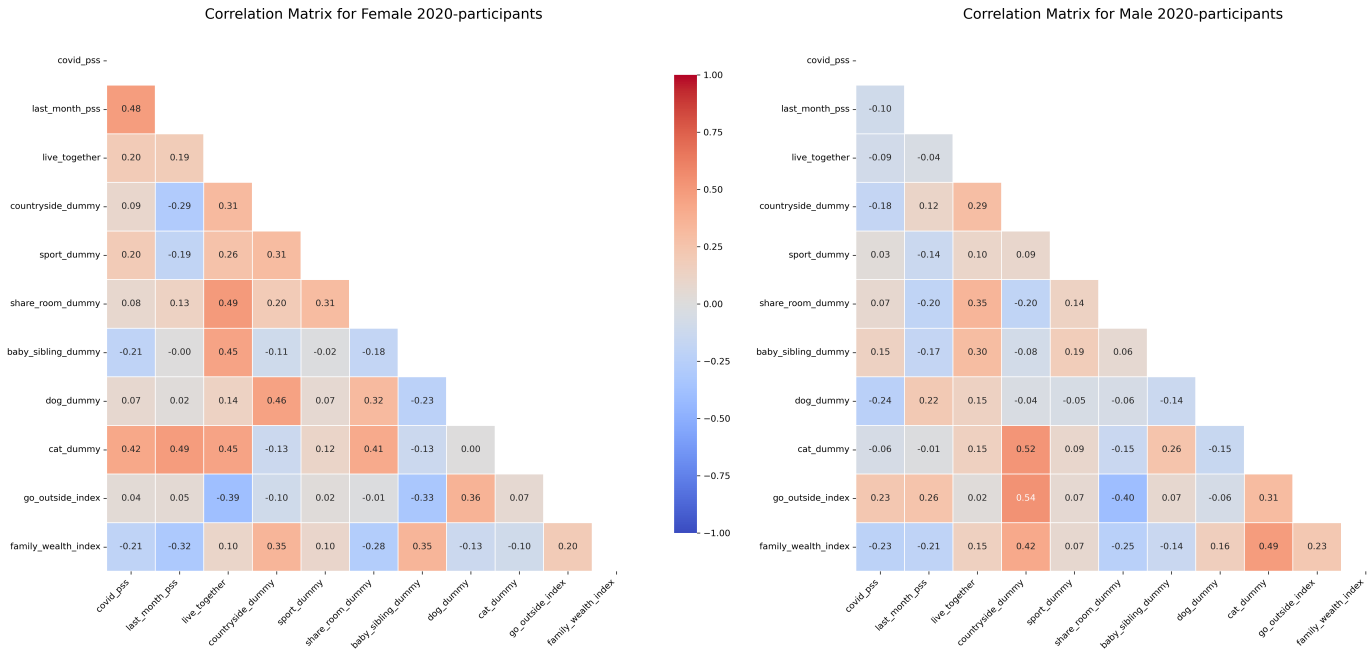
8.3 Additioanal materials

Рис. 2: Matrix of correlation for Controls by sex



Note: all participants aged 20 and above are included, except 2020-participants

Рис. 3: Correlation Matrix for Treatment by sex



Note: only 2020-participants are included

Список литературы

Ababkov, V., K. Barisnikov, O. Vorontzova-Wenger, I. Gorbunov, S. Kapranova, E. Pologaeva, and K. Stuklov (2016). Validation of the russian version of the questionnaire "scale of perceived stress–10". *Vestnik of Saint-Petersburg University. Education. 2*, 6–15.

Agnafors, S., M. Barmark, and G. Sydsjö (2020). Mental health and academic performance: A study on selection and causation effects from childhood to early adulthood. *Social Psychiatry and Psychiatric Epidemiology 56*(5), 857–866.

Beetz, A., K. Uvnäs-Moberg, H. Julius, and K. Kotrschal (2012). Psychosocial and psychophysiological effects of human–animal interactions: The possible role of oxytocin. *Frontiers in Psychology 3*.

Chumakov, E., N. Petrova, T. Mamatkhodjaeva, A. Ventriglio, D. Bhugra, and A. Molodynski (2021). Screening of minor psychiatric disorders and burnout among medical students in st. petersburg, russia: a descriptive study. *Middle East Current Psychiatry 28*(1), 38.

Cohen, S. (1988). *Perceived Stress in a Probability Sample of the United States*, pp. 31–67. Sage Publications.

Cohen, S., T. Kamarck, and R. Mermelstein (1983). A global measure of perceived stress. *Journal of Health and SocialBehavior 24*, 385–396.

Fomina, T. G., E. F. Filippova, A. V. Burmistrova-Savenkova, and V. I. Morosanova (2024). Stress in the educational environment and its impact on academic success and psychological wellbeing of students. *National Psychological Journal 19*(4), 148–160.

for Economic Co-operation, O. and Development (2016). How is learning time organised in primary and secondary education? Technical Report 38, OECD Publishing.

Greenberg, P. E., T. Sisitsky, R. C. Kessler, S. N. Finkelstein, E. R. Berndt, J. R. T. Davidson, J. C. Ballenger, and A. J. Fyer (1999). The economic burden of anxiety disorders in the 1990s. *The Journal of Clinical Psychiatry 60*(7), 427–435.

Halpern-Manners, A., L. Schnabel, E. M. Hernandez, J. L. Silberg, and L. J. Eaves (2016). The relationship between education and mental health: New evidence from a discordant twin study. *Social Forces 95*(1), 107–131.

Kaplan, D. S., R. X. Liu, and H. B. Kaplan (2005). School related stress in early adolescence and academic performance three years later: The conditional influence of self expectations. *Social Psychology of Education 8*(1), 3–17.

Kertes, D. A., J. Liu, N. J. Hall, N. A. Hadad, C. D. L. Wynne, and S. S. Bhatt (2017). Effect of pet dogs on children's perceived stress and cortisol stress response. *Social Development 26*(2), 382–401.

Kessler, R. and P. Greenberg (2002). The economic burden of anxiety and stress disorders. *Neuropsychopharmacology: The Fifth Generation of Progress 67*, 981–992.

Kostromina, S. N. and A. E. Pisarev (2017). Examination stress at unified state examination: Student destabilization or success factor? *Statistics and Economics* (3), 80–91.

Lisnyj, K. T., N. Gillani, D. L. Pearl, J. E. McWhirter, and A. Papadopoulos (2021). Factors associated with stress impacting academic success among post-secondary students: A systematic review. *Journal of American College Health 71*(3), 851–861.

Marakshina, J., T. Adamovich, G. Vasin, V. Ismatullina, M. Lobaskova, A. Malykh, P. Kolyasnikov, A. Tabueva, I. Zakharov, and S. Malykh (2024). Factor structure and psychometric properties of the perceived stress scale in russian adolescents. *Scientific Reports 14*(1), –.

Matyushkina, E. (2016). Academic stress of students with different forms of learning. *Консультативная психология и психотерапия 24*(2), 47–63.

Middlebrooks, J. S. and N. C. Audage (2008). The effects of childhood stress on health across the lifespan. *Centers for Disease Control and Prevention*.

Monroe, S. and G. Slavich (2016). Psychological stressors: Overview. *Stress: Concepts, Cognition, Emotion, and Behavior*, 109–115.

Pascoe, M. C., S. E. Hetrick, and A. G. Parker (2020). The impact of stress on students in secondary school and higher education. *International Journal of Adolescence and Youth 25*(1), 104–112.

Smith, E. and S. Kosslyn (2007). *Cognitive Psychology: Mind and Brain*. Pearson/Prentice Hall.

Symaniuk, E. E., L. V. Okonechnikova, and E. S. Malova (2019). Research of educational stress and its communication with achievement at students of senior classes. *Pedagogical Education in Russia* (8), 42–49.

Wälde, K. (2018). Stress and coping—an economic approach. In *CESifo Working Paper Series*, Number 6966.

Ye, B.-J., X.-H. Hu, and Z.-J. Hu (2014). The effect mechanism of perceived social support, coping efficacy and stressful life events on adolescents' academic achievement. *Journal of Psychological Science 37*(2), 342–348.

Ye, L., A. Posada, and Y. Liu (2019). A review on the relationship between chinese adolescents' stress and academic achievement: Stress and academic achievement. *New Directions for Child and Adolescent Development 2019*(163), 81–95.

Zhang, W. (2010). Adolescence's family financial difficulty and social adaptation: Coping efficacy of compensatory, mediation, and moderation effects. *Journal of Beijing Normal University*.

Zolotareva, A. (2023). Psychometric properties of the russian version of the perceived stress scale (pss-4, 10, 14). *Clinical Psychology and Special Education 12*(1), 18–42.

Сёмина, and Фёдорова (2023). Проявление академического стресса у студентов: теоретические аспекты исследования. *Северо-Кавказский психологический вестник 21*(3), 63–74.